

Empirical Benchmarking of Neural Text-to-Speech Architectures for Indic Languages: Does Explicit Devanagari Grapheme-to-Phoneme Conversion Still Matter?

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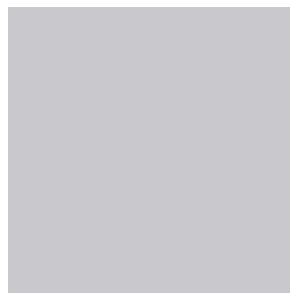
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1 Introduction

1.1 Background

Two Hindi words make the problem concrete. *Namak* (salt) is pronounced /nəmək/; *namkeen* (savoury) is /nəmki n/. Both are written in Devanagari with no vowel sign after the second consonant, yet one keeps a vowel there and the other drops it. The spelling does not tell you which.

This is schwa deletion. Devanagari is an abugida, so every consonant carries an unwritten inherent vowel, the schwa ə, and Hindi deletes many of them. The standard treatment is still the iterative right-to-left rule of Narasimhan, Sproat and Kiraz [13], which deletes a schwa lying between two consonants that are themselves flanked by vowels. It works well but not perfectly, with errors clustering at morpheme boundaries and in loanwords. Arora et al. [14] later recast the task as supervised sequence labelling and improved on it.

Synthesis itself has moved on entirely. Three architectures cover most of current practice. FastSpeech 2 [1] is a non-autoregressive transformer and it needs explicit durations. VITS [2] maps text to waveform through a variational autoencoder trained adversarially, and it learns its own alignment. Matcha-TTS [3] uses conditional flow matching to reach diffusion-level quality in far fewer steps. Where a spectrogram is produced, HiFi-GAN [4] turns it into audio.

Indic resources have improved too. They run from the IIT Madras consortium recordings [12] through AI4Bharat’s baselines [10] and IndicVoices-R [11] to MMS [9]. The MMS pretrained VITS checkpoints cover both my languages, so fine-tuning is now the realistic starting point. One further fact shapes this study. Marathi uses the same script as Hindi and shares most of its phoneme inventory [15, 16], but it does not apply Hindi’s medial schwa deletion rule.

1.2 Challenges

Devanagari underspecifies the vowel, so a phonemic front end must guess, and sometimes guesses wrong. Beyond that, published comparisons rarely hold anything constant: two systems typically differ in architecture, data, sampling rate, vocoder and front end at once, so no quality gap can be attributed to any one of them.

The metrics are easy to get wrong too. MCD that keeps the zeroth coefficient is partly a loudness measure [17]. Pitch error on a linear scale is dominated by the higher-pitched speaker. And word error rate means nothing on its own. You need the recogniser’s own error rate on natural audio, which is the floor, and it is rarely reported.

1.3 Research Gap

Strong single-system Indic results exist, as do architectural comparisons for English. What does not exist is a controlled measurement of how much the front end contributes, separated from the architecture, on a Devanagari language.

Three things are missing specifically. No study varies only the input representation while holding architecture, data, budget, vocoder and evaluation fixed. No study uses a same-script control, so a difference found on Hindi alone cannot distinguish "Devanagari is hard" from "schwa deletion is hard". And no study reports how the effect changes with data quantity, though a team with ten hours and a team with ten minutes face different questions.

1.4 Motivation

The practical question is whether to write a G2P front end at all. It is real work, and building mine took most of a month. If current architectures learn schwa deletion from data, that month was wasted and nobody else should spend it. If they do not, the field should stop defaulting to character-level input for Indic languages, which is presently common. Either answer is useful and neither has been published.

1.5 Problem Statement

Holding training budget, evaluation protocol and speaker fixed, does phonemic input from an explicit Devanagari G2P front end improve neural TTS quality in Hindi relative to raw grapheme input? Does the answer depend on the acoustic architecture? Does it survive as training data falls from nine hours to ten minutes? And using Marathi as a same-script control where medial schwa deletion does not apply, is any effect attributable to schwa deletion rather than to the script in general?

1.6 Research Objectives

- O1. Build a speaker-matched, loudness-normalised Hindi and Marathi corpus with frozen, checksummed splits that regenerate identically.
- O2. Implement a Devanagari G2P front end with schwa deletion as a separately switchable stage, and measure its accuracy against elicited speaker judgement before trusting it downstream.
- O3. Train FastSpeech 2, VITS and Matcha-TTS under a provably identical budget, on phonemic and on graphemic input.
- O4. Measure the phoneme-versus-grapheme effect across a nested data ladder from nine hours to ten minutes.
- O5. Repeat the ablation in Marathi to separate phonology from script.

- O6. Evaluate with verified objective metrics, predicted MOS, a human listening test, and real-time factor on consumer hardware.
- O7. Release the testbed, the checkpoints and a speaker-validated contested-schwa stress-test set.

2 Literature Survey

Architectures. FastSpeech 2 [1] (27M) predicts duration, pitch and energy explicitly then generates the spectrogram in parallel, which makes it fast but dependent on ground-truth durations from somewhere, traditionally an external aligner. VITS [2] (83M) collapses the pipeline into one model and learns alignment internally by monotonic alignment search, inherited from Glow-TTS [5]. Matcha-TTS [3] (18M) uses optimal-transport conditional flow matching to match diffusion quality in fewer steps, which makes it interesting at the low-data end of my ladder. StyleTTS 2 [6] reports stronger naturalness than any of them; I cut it for compute.

Vocoding. HiFi-GAN [4] remains the default. At around 14M parameters for the V1 generator it is cheap beside the acoustic models, which is why my design holds it fixed across the two systems needing one.

Alignment. The Montreal Forced Aligner [8] has been the conventional duration source for FastSpeech-family models and is what my original plan assumed. Badlani et al. [7] then showed that an alignment module trained jointly with the TTS model, without external supervision, matches or beats forced-aligner durations. That result changed my design; Section 4.5 explains why.

Indic resources. The consortium recordings [12] sit behind most Indian-language TTS work, including both corpora here. Kumar et al. [10] showed usable quality from modest per-language data. IndicVoices-R [11] scaled that up for synthesis. MMS [9] supplies the checkpoint my VITS condition initialises from. Between them these papers establish that Indic TTS works and how well it works. The contribution of the front end is not separated out in any of them, and that is what I am measuring.

Evaluation. Mel-cepstral distortion [17] is the standard spectral measure; pitch and aperiodicity usually come from WORLD [18]; intelligibility is increasingly assessed by transcribing synthesised audio with Whisper [19]. Listening tests are slow, so automatic MOS predictors such as UTMOS [20] and NISQA [21] have become common. Both were trained largely on English and Japanese. I will measure how well they correlate with my own human ratings and report that as a result. Loudness normalisation follows ITU-R BS.1770 [22].

3 Experimental Setup and Work Done Till Now

Corpora and standardisation. I use the Hindi and Marathi corpora from the SPRING Lab IndicTTS collection. Neither dataset card states hours, speaker count or sampling rate, so I measured all three. An early estimate from a 50-utterance sample proved 19% low for Hindi. Every utterance then goes through the same chain. It is downmixed to mono, silence-trimmed at 40 dB below peak with 50 ms of padding kept, and normalised to -23 LUFS [22]. From there it is resampled twice, to 22.05 kHz for the master and separately to 16 kHz for the VITS arm, and peak-guarded at 0.99. Two details in that chain are deliberate. Trimming happens at the 48 kHz source, so both outputs share their boundaries to the sample. And the 16 kHz copy comes from that same gained source, not from the master. Cascading two resamples would put a measurably different signal in front of VITS than the other architectures see, and that difference would show up in the results looking like an architectural effect.

G2P front end. Four separable stages, arranged so the phonological rule can be switched without touching anything else. Normalisation handles NFC, nukta composition, corpus spelling repair, digit expansion and punctuation folding. Segmentation makes every inherent schwa explicit and tags it as unwritten. Schwa deletion applies the rule from [13] plus obligatory word-final deletion plus a lexicon override; Marathi runs identical code with medial deletion disabled, and that single flag is the whole control condition. Finally everything maps into one IPA inventory shared by both languages. A parallel grapheme path reuses the identical normalisation chain, so the ablation measures phonology rather than preprocessing.

Gold pronunciations without circularity. I wrote the schwa rule, so gold forms I also wrote would measure only my own consistency. They come from native speakers instead, through a sheet that never shows the rule’s prediction. Each word appears with its candidate vowels spelled out in numbered slots, as in $n(a1)m(a2)kiin(a3)$. Speakers mark each one yes, no or variable. They do not write IPA, because transcription by non-phoneticians adds noise that has nothing to do with the question. Their answers pass through the same converter to produce IPA. Items where speakers disagree are marked variable, excluded from the accuracy denominator and reported as a variation rate.

Metrics. MCD uses DTW alignment over 24 coefficients, excluding c_0 . The alignment matters: a model with a duration predictor differs systematically in timing from an end-to-end one, and frame-aligned comparison would charge that timing difference as spectral error. Prosody is reported as two separate numbers, $\log-F_0$ RMSE over commonly voiced frames and voiced/unvoiced error rate. A system with correct voicing and wrong pitch has a different defect from one with the reverse.

Design and reproducibility. There are seventeen configurations (Table 4). Every

acoustic run shares 100,000 steps, 12,000 mel frames per batch, a learning rate of 2×10^{-4} with inverse-square-root decay after 4,000 warmup steps, fp16, and gradient clipping at 1.0. That sentence is the study’s central claim, so an assertion enforces it. I am not relying on remembering it. One YAML file defines one run, and it carries a hash over every field that affects the trained model. A second assertion refuses two runs that share a hash. The splits are locked with a SHA-256 per file. Preparation, G2P and evaluation all run locally on an Apple M5. Training is planned on free-tier GPU resources, capped at 30 GPU-hours a week in 12-hour sessions, which is where the checkpoint-resume requirement comes from.

4 Results and Discussion

Nothing has been trained yet, so what follows are properties of the corpus, the front end and the measuring apparatus. Each had to be established before any model was trained. Several turned out to be findings in their own right.

4.1 Corpus Profile

Neither dataset labels its speakers; there is a binary field with no explanation. Hindi and Marathi have to be matched on speaker characteristics, so I recovered the labels from median F_0 over 25 sampled utterances per speaker. It separates them cleanly, and identically, in both languages (Table 1). Every gender label here rests on that measurement. The dataset itself says nothing about it.

Table 1: Measured corpus properties. Gender is inferred from median F_0 .

	Hindi		Marathi	
	Speaker 0	Speaker 1	Speaker 0	Speaker 1
Utterances	5,983	5,842	5,362	5,577
Hours (raw)	13.10	13.13	11.49	10.88
Median F_0 (Hz)	198.9	108.8	237.2	125.7
Inferred gender	female	male	female	male
Corpus total	11,825 utts, 26.23 h, 48 kHz		10,939 utts, 22.37 h, 48 kHz	

I selected the male speaker in both. The hours are near identical either way, so pitch decided it. The two male voices sit 17 Hz apart across the two languages. The female pair sit 38 Hz apart. Picking the men leaves less acoustic difference between the arms that has nothing to do with schwa deletion.

I checked transcript quality with characters per second of audio. A transcript that belongs to different audio shows up as an outlier on that measure. Hindi’s median is 12.52 (p1 9.16, p99 15.77) and Marathi’s is 9.10 (6.64, 11.70). Nothing in either falls outside 0.4 to 2.0 times the median, so there is nothing to clean. The longest Hindi utterance

runs 127.7 s with 1,600 characters. I assumed it was corrupt. That works out at 12.5 characters per second, which is exactly the corpus median.

4.2 Standardisation and Splits

Trimming removed 5.5% of Hindi and 8.4% of Marathi. Marathi carried noticeably more dead air. That had a consequence I had not planned. After trimming, and after the frozen test and development sets are taken out, only 9.53 h and 9.16 h of training audio survive. My ladder was going to start at ten hours. It now starts at nine, because matching the rungs across the two languages matters more than a round number. An unmatched top rung would drop a data-quantity difference into the control comparison, and that is the one place it must not be.

Table 2: Standardised corpora (male speaker, 1–15 s, −23 LUFS) and the nested data ladder in utterances per rung.

	Hindi	Marathi	Rung	Hindi	Marathi
Utterances retained	5,485	5,559	9 h	4,793	5,071
Hours	10.289	9.884	5 h	2,668	2,810
Peak-limited	1	0	1 h	541	562
Test / dev	300 / 100	300 / 100	30 min	265	279
Train (hours)	5,085 (9.525)	5,159 (9.160)	10 min	86	90

Spot-checking confirmed exactly −23.00 LUFS at both rates. The two copies agree on duration to under a millisecond, the peaks stay under the ceiling, and the edge padding is there. The peak guard fired once. So it is not dead code.

Split membership comes from a salted SHA-256 of the utterance identifier. A seeded shuffle, the usual choice, depends on the order the manifest happened to be written in, so re-running preparation elsewhere can hand you a quietly different test set. Hashing the identifier makes membership a property of the utterance, and rebuilding reproduces the test split byte for byte. The ladder takes prefixes of that one ordering. Sampling each rung independently would let sampling variance look like a data-quantity effect, which is the effect the ladder exists to measure. I verified that every rung is a strict superset of the one below, that the three splits are pairwise disjoint, and that no rung reaches outside training.

4.3 G2P Coverage

I developed the front end against 49 hand-picked words and it handled all of them. Then I ran it over all 11,044 corpus transcripts. 4.1% of Hindi phone tokens and 2.7% of Marathi had no mapping at all. Both are now zero (Table 3). Getting there meant fixing seven things a curated test set was never going to find. Nine Hindi tokens carried a nukta on a consonant that has no nukta form, and the stray mark also broke the parse of

the next character. 199 Marathi tokens were the letter RRA. There were keyboard slips producing an invalid candra-O sequence, and a colon typed where a visarga belonged. The OM ligature is a syllable written as a single letter. Southern short vowels appear that neither language distinguishes. And some vowel signs had no consonant to attach to.

Table 3: G2P coverage over both corpora after correction.

	Hindi	Marathi
Word tokens	99,958	55,492
Phone tokens	494,863	375,684
Unknown symbols	0	0
Inventory used / declared	72 / 75	67 / 75

Marathi’s inventory is a strict subset of Hindi’s. It lacks only five marginal or Perso-Arabic phones, so the two arms share one symbol table with no Marathi-only embedding rows. One weakness is worth recording here. Three Marathi phones occur fewer than three times in the whole corpus, so those embedding rows will get essentially no gradient.

4.4 Metric Verification

I checked both metrics against answers I could derive on paper.

MCD proved amplitude invariant. Scaling the input by 0.25 changes it by under 10^{-6} dB. That is the property that confirms the implementation. Scaling multiplies the power spectrum by a constant, which is a constant decibel offset across every mel band, and that lands entirely in the zeroth DCT coefficient. Dropping c_0 therefore makes the measure blind to gain. If that test ever failed, every MCD figure here would be partly a loudness measurement, and nothing in the results would show it.

Log- F_0 RMSE reports $\ln(1.1)$ for a 10% pitch error at 100 Hz and again at 300 Hz, exact to 10^{-12} . A linear RMSE would call the same perceptual error 10 Hz on one voice and 30 Hz on another. Two further guards are tested. Unequal-length F_0 tracks raise rather than truncate, because truncation would favour systems that predict shorter durations. And a comparison with no commonly voiced frames returns NaN, not zero. The suite stands at 78 tests.

4.5 Three Defects Found

Each would have corrupted results without any visible symptom, and two were mine.

Punctuation was rewriting words. I kept punctuation as a prosodic token, since it is the only phrasing cue the text carries. Leaving it attached through the schwa stage breaks the rule twice over. A trailing comma means the word-final schwa is no longer final, so it survives. Its survival then supplies the right-hand vowel that lets the schwa

to its left delete. *Namak*, /nəmək/, came out /nəmkə/, a different word. 16.5% of Hindi word tokens and 14.9% of Marathi carry punctuation. The damage would have fallen on the phonemic arm alone, because the grapheme path never touches the schwa stage. My headline comparison would have been partly measuring this bug. Punctuation is now stripped before the schwa stage and restored after. There is an invariance test across nine words and four marks in both languages.

A run was in the plan twice. The design had 19 runs. The ladder’s nine-hour rung was a second copy of the main run under a different identifier. That is ten GPU-hours for a number I already had. It stayed invisible because the run identifier sat inside the configuration hash, which by construction makes two identical runs look distinct. The identifier is now excluded, and an assertion refuses any two runs that share a settings hash.

The elicitation sheet gave away its own answers. Its worked examples were themselves items from the 49-word test set. So every participant was handed the answers to three of the words they were about to be tested on. I noticed only because someone reading the sheet asked about one. Examples now come from outside the set, enforced by a check at generation time.

4.6 Two Design Decisions

Forced alignment dropped. My plan took FastSpeech 2’s durations from MFA [8]. I have replaced that with internal unsupervised alignment [7], for three reasons. The first is that it was an uncontrolled asymmetry. VITS and Matcha both learn alignment internally [5], so externally supervised durations would give one architecture information the other two never receive, inside a comparison whose subject is the architecture. The second is that it was asymmetric across languages as well. MFA publishes a pretrained Hindi model and no Marathi one, which puts that difference inside the control comparison. The third is that MFA would not run in my environment. The package installs and then fails on Kaldi bindings, which have no wheel for my architecture.

One limitation follows from the change. Internal alignment is weaker when data is very scarce, so the lowest ladder rungs may suffer more than the top ones. I will measure that as a property of the ladder and report it.

Sampling rate. The master is 22.05 kHz, as the synopsis specified. But VITS initialises from an MMS checkpoint [9] that is natively 16 kHz, so its output carries nothing above 8 kHz. Full-band MCD would then charge it for missing bandwidth rather than for worse modelling. MCD is reported in two columns, full band and band-limited to 8 kHz, and the band-limited column is the headline. Listening stimuli are low-passed to 8 kHz for every system, so raters hear equal bandwidth. ASR-WER needs no mitigation at all, since Whisper resamples to 16 kHz anyway [19].

4.7 Compute Requirements

Table 4: The experimental matrix. All acoustic runs share one budget, enforced by assertion.

Run	Architecture	Lang.	Input	Rate (kHz)	GPU-h
r01 / r02 / r03	FS2 / VITS / Matcha	hi	phoneme	22.05 / 16 / 22.05	36
r04 / r05	FS2 / VITS	hi	grapheme	22.05 / 16	16
r06	HiFi-GAN	hi	n/a	22.05	10
r07-r10	FastSpeech 2	hi	phoneme, ladder	22.05	20
r11-r14	VITS	hi	phoneme, ladder	16	20
r15 / r16	FS2 / VITS	mr	phoneme	22.05 / 16	24
r17	HiFi-GAN	mr	n/a	22.05	10
Total (17 runs)					136

Table 4 totals 136 GPU-hours at one seed per configuration. On free-tier resources capped at 30 hours a week, that is 4.53 weeks of training alone. One seed is not really enough. My rule is that no difference smaller than the measured seed variance gets reported as a finding, and with a single seed everywhere that floor rests on replicating one cell and generalising from it. Three seeds throughout would take the requirement to 368 GPU-hours.

Institutional GPU access would mostly buy statistical strength. On a DGX A100 the single-seed campaign finishes in roughly 1.6 days, and three-seed replication in about 4.4. The calendar saving is around four weeks out of thirteen, since evaluation and the listening study wait on people. The larger gain is in what I can claim. Every comparison moves from “these numbers differ” to “these differ by more than run-to-run noise”.

5 Conclusion

Preparation is done. Two Devanagari corpora totalling 48.6 hours have been profiled, standardised, and split into frozen checksummed partitions that regenerate identically. Matching them on speaker characteristics meant recovering those characteristics acoustically first, because the datasets do not record them. The G2P front end is built, with schwa deletion isolated as a switchable stage, and it covers both corpora completely. The instrument for validating it against native-speaker judgement is waiting on participants. Both objective metrics are checked against analytic answers. All 17 runs are specified, and the shared budget is enforced by assertion.

Three defects turned up along the way and none of them announced itself. The worst affected 16.5% of Hindi word tokens and would have degraded only the phonemic arm, which is the half of the comparison this study rests on. All three were found the same way: by running the code over the real corpus, and by watching someone try to use the elicitation sheet. The 78 unit tests caught none of them. I will pilot the listening study

on two or three people before recruiting twenty.

Remaining immediately: collecting elicitation sheets, confirming the GPU execution path end to end, and writing the training wrappers.

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